

Indian physicist Sir Chandrasekhara Venkata Raman was awarded the 1930 Nobel Prize in Physics for his work on the scattering of light and for the discovery of the effect named after him. He also promoted the growth of science in India.

MILESTONES

TEENAGE PRODIGY

Gains a degree from the University of Madras in 1904, earning his master's degree 3 years later.

NEW DIRECTION

Leaves accountancy to become professor of physics at the University of Calcutta in 1917.

PROMOTES SCIENCE

Sets up the *Indian Journal of Physics* in 1926; founds the Indian Academy of Sciences in 1934.

LIGHT EXPERIMENT

In 1928, publishes leading research into the behavior of light particles; this wins him a Nobel Prize in 1930.

RESEARCH INSTITUTE

Becomes the founder and first director of the Raman Research Institute in 1948, based in Bangalore.

Born in Madras State, now Tamil Nadu, in southern India, C. V. Raman was just 14 when he enrolled at the University of Madras. After graduating, he went on to gain a master's degree. In 1907, he became an accountant but continued with scientific research in his spare time. Raman established a reputation as a highly able physicist and in 1917 was made professor of physics at the University of Calcutta, where he conducted extensive research into acoustics and optics.

Seeing the light

Raman's voyage to London in 1921 inspired him to investigate why the sea appears blue. He discovered that when a beam of light shines on a transparent substance, some of that light is scattered and a small amount alters in wavelength. This prompted his extensive series of experiments to measure how light was scattered by liquids, solids, and gases and led to his discovery of the Raman effect. This effect occurs when photons (the fundamental particle of light) either take up energy from or lose energy to a molecule. They then scatter with increased or decreased energy and a different wavelength. This explained why water molecules in the sea scatter white sunlight and the resulting light falls mainly in the blue part of the light spectrum. The Raman effect was considered by many to confirm Einstein's quantum theory, and it remains a key technique in substance analysis. In 1930, Raman became the first Asian to be awarded a Nobel Prize. He devoted the rest of his career to teaching students, founding academic publications, and setting up research institutions.

Raman, pictured here at his Raman Research Institute, discovered the Raman effect—which is now the basis of a laboratory technique called Raman spectroscopy.

